

PRAVEEN KUMAR

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PROFESSIONAL SUMMARY

Senior Backend Engineer / Backend Architect with 12+ years of experience designing, building, and optimizing scalable backend systems and high-traffic applications serving millions of users. Strong hands-on expertise in **PHP, Laravel, Node.js, Python, REST APIs, GraphQL, MySQL, MongoDB, Redis, AWS, and Microservices**, with 5 years of technical leadership experience. Proven experience in backend architecture, API development, performance optimization, database tuning, system modernization, production troubleshooting, and mentoring engineering teams. Additionally, hands-on experience developing **Generative AI, RAG, Agentic AI, and LLM-based applications** through independent projects using modern AI frameworks and tools.

CORE TECHNICAL SKILLS

- **Backend:** PHP, Laravel, Node.js, Python, FastAPI
- **Architecture:** System Design, Scalable Systems, REST APIs, GraphQL, Microservices
- **Databases:** MySQL, MongoDB, Redis, Vector Databases
- **Cloud & DevOps:** AWS, Docker, CI/CD
- **Generative AI:** LLM APIs, RAG, Agentic AI, LangChain, LangGraph, CrewAI, MCP, Google ADK, OpenAI, Anthropic
- **AI Engineering:** Prompt Engineering, AI Agents, LLM Evaluation, Guardrails, Vector Search
- **Architecture:** System Design, Scalable Systems, REST APIs, GraphQL, Microservices
- **Leadership:** Technical Leadership, Mentoring, Code Reviews, Technical Design, Cross-functional Collaboration

PROFESSIONAL EXPERIENCE

Senior Project Lead

Aug 2019 – Sep 2024

Matrimony.com Limited — Chennai, India

- Provided technical leadership for backend systems supporting millions of users, driving scalable architecture and engineering best practices.
- Led backend development while mentoring engineers through technical design discussions, code reviews, troubleshooting, and hands-on development.
- Collaborated with Product, QA, and Infrastructure teams to deliver scalable, reliable, and maintainable backend solutions.
- Participated in technical interviews and contributed to maintaining high engineering standards.

Key Engineering Contributions:

- Improved backend performance by approximately 30% through API optimization, MySQL query tuning, and Redis caching.
- Designed and developed scalable backend services for authentication and photo management modules supporting millions of users.
- Contributed to the migration of authentication and photo management modules from PHP to Node.js using GraphQL APIs.
- Developed reusable backend components to improve development efficiency and code consistency.
- Contributed to production stability through deployment support, incident troubleshooting, and performance optimization.
- Identified architectural bottlenecks and implemented scalable solutions to improve system reliability and maintainability.

Technologies: PHP, Laravel, Node.js, Python, GraphQL, REST APIs, MySQL, MongoDB, Redis, AWS, Microservices, Agile

Senior Web Developer

Jul 2018 – Aug 2019

Innoblitz Technologies — Chennai, India

- Delivered backend feature enhancements improving delivery efficiency by 20%
- Optimized database queries improving response performance by 40%
- Collaborated with product and QA teams for feature rollout and defect resolution

Lead Web Developer

Jun 2015 – Jul 2018

Times Partner Services Pvt Ltd — Chennai, India

- Led backend development for enterprise web applications
- Maintained 99.9% system uptime through performance and stability improvements
- Reduced development cycle time by 15% through reusable service components

Earlier Experience

Web Developer – Dot Infotech (Aug 2012 – May 2015)

Delivered web application modules, improved database performance, and supported full development lifecycle.

TECHNICAL PROJECTS

Agentic RAG Assistant

GitHub: <https://github.com/praveenkhumar/agentic-rag-assistant>

Built a FastAPI service orchestrated with LangGraph (branching retrieve-decide-tool-call-generate-guardrail workflow) backed by a locally persistent Chroma vector store, with an MCP stdio server exposing calculator and web-search tools for when local context is insufficient. Added LangSmith run tracing, inspectable guardrails (PII detection, heuristic groundedness, scope refusal), and an evaluation harness measuring retrieval hit@k and LLM-judged answer groundedness, deployed via Docker and Render with startup-time index rebuilding for ephemeral environments.

Tech: Python, FastAPI, LangGraph, LangChain, Chroma, MCP, LangSmith, Docker, pytest

Multi-Agent Task System, CrewAI

GitHub: <https://github.com/praveenkhumar/multi-agent-task-system>

Built a CrewAI research-to-report pipeline with Researcher, Writer, and Reviewer agents; the Researcher uses live Tavily search, with the LLM provider configurable between OpenAI and Anthropic. Implemented a governed reject-and-revise workflow (maximum one revision) with structured handoff tracing, five evaluation criteria validated across five topics using deterministic fixtures, and a FastAPI /run endpoint returning the final report, evaluation scores, and full execution trace, deployed via a Render Blueprint with a /health check.

Tech: Python, CrewAI, FastAPI, Tavily API, OpenAI/Anthropic, Render

EDUCATION

Master of Science – Information Technology

University of Madras, Chennai

Bachelor of Computer Applications

D.G. Vaishnav College, Chennai

CERTIFICATIONS

- Generative AI (GUVI - HCL), Nov 2025
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents (Udemy), Sep 2025
- Machine Learning, Data Science & AI Engineering with Python (Udemy), Apr 2025
- NodeJS - The Complete Guide (MVC, REST APIs, GraphQL, Deno) (Udemy), Nov 2024
- Laravel - For Beginner to Advanced (Udemy), Nov 2024